A CASE-STUDY BASED REPORT

ON

SOIL FERTILITY CLASSIFICATION USING DECISION TREE AND RANDOM FOREST MACHINE LEARNING

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Abstract

Soil fertility is an important factor affecting agricultural productivity, crop growth, and sustainable land management. Assessing soil fertility requires considering several nutrient and chemical properties together, rather than relying on a single measurement. This project presents a machine-learning-based approach for classifying soil fertility from numerical soil measurements. The system uses 12 input parameters: nitrogen (N), phosphorus (P), potassium (K), pH, electrical conductivity (EC), organic carbon (OC), sulphur (S), zinc (Zn), iron (Fe), copper (Cu), manganese (Mn), and boron (B).

The project workflow includes loading and preparing the soil dataset, removing duplicate records, exploring the data, separating input features from the target fertility class, training classification models, validating and tuning the models, and evaluating their performance. Two algorithms are considered: Decision Tree and Random Forest. Five-fold cross-validation and GridSearchCV are used as part of the validation and hyperparameter tuning process. The evaluation framework includes accuracy, precision, recall, F1-score, and a confusion matrix.

According to the project presentation, Random Forest achieved the best reported test accuracy of 89.20% and was selected as the final model. The trained model is saved using Joblib and connected to a Gradio interface, where a user can enter the 12 soil parameter values and view the predicted fertility class along with confidence or class probabilities. The project demonstrates how machine learning can support consistent, data-driven soil classification. Its predictions are intended as preliminary decision support and do not replace laboratory soil testing or professional agricultural advice.

1. Introduction

1.1 Background

Soil is a fundamental resource for agriculture because it supplies nutrients, supports plant roots, stores water, and contributes to crop development. Soil fertility depends on a combination of physical, chemical, and nutrient-related conditions. Nutrients such as nitrogen, phosphorus, and potassium are important for plant growth, while properties such as pH, electrical conductivity, organic carbon, and micronutrients also contribute to the overall condition of the soil.

A soil sample therefore cannot always be assessed adequately by considering one parameter in isolation. The values of several measurements must be interpreted together. Machine learning provides a way to learn relationships from labelled examples and use those learned patterns to classify new samples. This project applies supervised classification methods to soil nutrient and chemical measurements to predict a corresponding fertility class.

1.2 Motivation for the Project

The motivation for this project is to explore whether soil measurements can be used to produce a quick, consistent, data-driven fertility classification. Traditional soil assessment can involve laboratory testing and interpretation of several parameters. Automating the classification stage may reduce repetitive manual analysis and make a preliminary result easier to obtain.

Decision Tree and Random Forest were selected because both are tree-based classification approaches. A Decision Tree offers a rule-like structure that is relatively easy to explain, while Random Forest combines predictions from multiple trees and can provide feature-importance information. Comparing the two models allows the project to examine their performance on the prepared soil dataset.

1.3 Overview of the Work Carried Out

The project follows an end-to-end machine-learning workflow: dataset loading, data cleaning, exploratory data analysis (EDA), feature and target separation, model development, validation and hyperparameter tuning, model evaluation, model saving, and prediction through a user interface.

1. Load the soil fertility dataset and inspect its structure.
2. Check data quality and remove duplicate records.
3. Explore the soil parameters and target fertility class.
4. Train Decision Tree and Random Forest classifiers.
5. Use five-fold cross-validation and GridSearchCV for validation and tuning.
6. Compare model performance using accuracy, precision, recall, F1-score, and a confusion matrix.
7. Save the selected model using Joblib and provide a Gradio interface for new-sample prediction.

The supplied project presentation identifies Random Forest as the final model, with a reported test accuracy of 89.20%. Inspection of the provided CSV shows 1,288 rows and 13 columns (12 input parameters and the fertility target). The source of the dataset and the meanings of target labels 0, 1, and 2 are not documented in the supplied files. The train-test split ratio and Decision Tree test accuracy should be verified from the implementation.

1.4 Scope and Responsible-Use Statement

The scope of the project is to classify soil fertility using 12 numerical nutrient and chemical parameters. It includes preprocessing, EDA, model training, validation, comparison, model saving, and a simple prediction interface. The interface is intended to demonstrate how a trained classifier can be used with a new soil sample.

The output depends on the quality, representativeness, and labelling of the dataset used for training. A model prediction should not be treated as a replacement for laboratory testing, local agronomic guidance, or expert interpretation. Users should enter measurements in the units and ranges expected by the training data. The system's output is a model-generated classification, not a guarantee of crop performance or a recommendation to apply a particular fertilizer.

2. Problem Statement

2.1 Case-Study Scenario

Consider a farmer, soil-testing laboratory, agricultural advisor, or researcher who has measurements for a soil sample. The measurements may include major nutrients, micronutrients, pH, electrical conductivity, and organic carbon. Interpreting these values together to assign a fertility category can require time and familiarity with soil assessment.

The proposed system uses examples from a labelled soil dataset to learn patterns between the 12 measurements and the fertility class. Once trained, the model can accept a new set of measurements through the Gradio interface and return a predicted class with confidence or class-probability information, where supported by the implementation.

2.2 Central Case-Study Question

Can a supervised machine-learning model classify soil fertility from 12 nutrient and chemical measurements, and does Random Forest provide better predictive performance than a Decision Tree on the dataset used in this project?

2.3 Formal Problem Definition

Let the input feature vector for one soil sample be:

$$\mathbf{X = [N, P, K, pH, EC, OC, S, Zn, Fe, Cu, Mn, B]}$$

Let Y represent the labelled soil fertility class.

The classification task is to learn a mapping f(X) → Y from the training examples. For a new soil sample, the trained model uses its 12 input measurements to estimate the corresponding fertility class. If the selected model and interface expose class probabilities, those values may also be displayed as supplementary information.

The exact names and meanings of the fertility classes are not stated in the supplied presentation. They should be copied from the target column in the actual dataset rather than assumed or invented.

2.4 Objectives that Guide the Study

8. Prepare and inspect the soil fertility dataset.
9. Check data quality, including missing values and duplicate records.
10. Explore the distributions of soil parameters and the target class.
11. Use the 12 soil parameters as model inputs and fertility class as the target.
12. Develop Decision Tree and Random Forest classifiers.
13. Apply five-fold cross-validation and GridSearchCV for validation and hyperparameter tuning.
14. Evaluate and compare models using accuracy, precision, recall, F1-score, and a confusion matrix.
15. Select the better-performing model based on the reported evaluation.
16. Save the selected model using Joblib.
17. Provide a Gradio interface for prediction on new soil samples.

2.5 Assumptions and Limitations

Assumptions

- The dataset contains soil measurements and a corresponding fertility-class label.
- The 12 input parameters are available for both training examples and new samples.
- The labels in the dataset are suitable for supervised classification.
- New input values are provided in the same format and units as the training data.

Limitations

- The provided CSV contains 1,288 records and 13 columns. It contains 12 numerical input features and a target column named fertility. The target has values 0, 1, and 2; the meaning of each label and the original data source are not documented in the supplied files.
- The exact test accuracy of the Decision Tree model is not reported in the supplied presentation.
- Prediction quality depends on dataset quality, coverage, and label correctness.
- The presentation does not specify the exact train-test split ratio or the full hyperparameter search space.
- A predicted class does not replace laboratory testing or advice from qualified agricultural professionals.
- The presentation does not provide detailed per-class metrics or a numerical confusion matrix.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The provided dataset file is named “Soil Fertility Data (Modified Data).csv”. It contains 1,288 rows and 13 columns: 12 numerical input features and one target column, fertility. The input features are N, P, K, ph, ec, oc, S, zn, fe, cu, Mn, and B. The target values are 0, 1, and 2. The source of the dataset, collection period, measurement units, and the semantic names for the three target classes are not specified in the supplied files and should be confirmed before final submission.

Parameter	Meaning / description	Role
N	Nitrogen measurement	Input feature
P	Phosphorus measurement	Input feature
K	Potassium measurement	Input feature
pH	Soil acidity/alkalinity measurement	Input feature
EC	Electrical conductivity	Input feature
OC	Organic carbon	Input feature
S	Sulphur measurement	Input feature
Zn	Zinc measurement	Input feature
Fe	Iron measurement	Input feature
Cu	Copper measurement	Input feature
Mn	Manganese measurement	Input feature
B	Boron measurement	Input feature
Fertility class	Label assigned to each soil sample; exact class names must match the dataset	Target

Dataset inspection summary

Check	Observed value	Interpretation
Dataset shape	1,288 rows × 13 columns	12 inputs + 1 target
Missing values	0 across all columns	No missing values found in this CSV
Duplicate rows	408	Full-row duplicates before cleaning
Rows after duplicate removal	880	If all duplicate rows are removed
fertility = 0	401 before / 401 after	Encoded class; semantic label not supplied
fertility = 1	440 before / 440 after	Encoded class; semantic label not supplied
fertility = 2	447 before / 39 after	Large drop after deduplication; investigate

3.2 Purpose of EDA

Exploratory Data Analysis is used to understand the dataset before model training. It helps identify the data structure, the distribution and scale of the input variables, the balance of the target classes, and possible data-quality problems. EDA can also help reveal unusual values or relationships that may affect the models.

- Inspect the number of rows and columns, data types, and sample records.
- Check missing values in each column.
- Identify and count duplicate records.
- Review descriptive statistics for the 12 numerical parameters.
- Examine the distribution of the fertility target classes.
- Use appropriate plots, such as histograms, box plots, class-count plots, or a correlation heatmap, if implemented.
- Look for data-quality issues and possible leakage from columns that would not be available at prediction time.

3.3 Preparation Steps Actually Performed

Direct inspection of the provided CSV found no missing values in any of the 13 columns and 408 duplicate rows when comparing complete records. Removing all duplicate rows leaves 880 records. This step has a substantial effect on the target distribution: before deduplication, fertility labels 0, 1, and 2 occur 401, 440, and 447 times; after deduplication, they occur 401, 440, and 39 times. The large reduction in class 2 should be investigated before training because it may indicate repeated records or a dataset construction issue. The presentation specifies duplicate removal, but the exact preprocessing used for the reported 89.20% accuracy should be checked.

18. Dataset loading: load the soil dataset into the Python environment.
19. Data-quality checks: inspect missing values and duplicate records.
20. Duplicate handling: remove duplicate records as stated in the project workflow.
21. Feature and target split: use N, P, K, pH, EC, OC, S, Zn, Fe, Cu, Mn, and B as inputs and fertility class as the target.
22. Model validation: use five-fold cross-validation and GridSearchCV as described in the presentation.
23. Model persistence: save the selected trained model with Joblib for use by the prediction interface.

Important reporting note: include any additional operations—such as missing-value imputation, scaling, encoding, outlier treatment, or a particular train-test split ratio—only if they appear in the implementation actually used.

3.4 EDA Exploration – Implementation

The EDA stage should examine the structure and distribution of the input parameters and the fertility target. Descriptive statistics can help show the central tendency and spread of each parameter. Class-count plots can show whether the target categories are balanced. Histograms and box plots can help reveal skewed distributions and possible outliers, while a correlation heatmap can summarize pairwise linear relationships among numerical features.

EDA on the provided CSV shows 1,288 rows, 13 columns, no missing values, and 408 duplicate rows. The fertility target is encoded as 0, 1, and 2, with counts of 401, 440, and 447 before duplicate removal. After removing duplicate rows, 880 records remain and the class counts become 401, 440, and 39. This indicates a pronounced class imbalance after deduplication, so the preprocessing decision should be reviewed and class-wise metrics should be considered. The CSV alone does not establish the meanings of the target labels or the real-world significance of any individual feature.

3.5 Summary of EDA Findings

The provided dataset contains 12 numerical input parameters and a fertility target with three encoded values (0, 1, and 2). There are no missing values in the supplied CSV. However, 408 duplicate rows are present. Removing duplicates reduces the dataset from 1,288 to 880 records and reduces the count for target label 2 from 447 to 39, so the impact of deduplication must be examined carefully. The label meanings and dataset source remain unconfirmed. These observations should guide model validation and reporting.

4. Algorithm Selection and Implementation

4.1 Decision Tree Classifier

A Decision Tree is a supervised learning algorithm that classifies examples through a sequence of feature-based decisions. At each internal node, the model selects a split based on a feature and threshold or category. Following the branches leads to a leaf node containing the predicted class. For this project, the tree learns decision rules from the 12 soil parameters to predict the fertility label.

A major advantage of a Decision Tree is interpretability: its structure can be inspected as a set of decision rules. It also provides a useful baseline for comparing a single tree with an ensemble method. However, a single tree may be sensitive to the training data and can overfit if its complexity is not controlled. The presentation includes Decision Tree in the comparison but does not provide its exact test accuracy.

4.2 Random Forest Classifier

Random Forest is an ensemble-learning algorithm that combines many Decision Trees. The trees are trained using randomized samples and feature selection, and their outputs are combined to produce the final classification. By aggregating multiple trees, Random Forest can reduce the instability of a single tree and often performs well on tabular data.

Random Forest was selected as the final model in this project because it achieved the highest reported test accuracy. The presentation reports a test accuracy of 89.20%. Random Forest can also provide feature-importance estimates, which may help examine how the model uses the input parameters. Feature importance should be interpreted as a property of the trained model, not as proof that a feature causes soil fertility.

4.3 Validation, Hyperparameter Tuning and Evaluation

The project uses five-fold cross-validation and GridSearchCV as part of model validation and hyperparameter tuning. In five-fold cross-validation, the training data is divided into five folds. The model is trained and validated repeatedly so that each fold serves as the validation fold once. GridSearchCV evaluates specified hyperparameter combinations using cross-validation and identifies a configuration according to the chosen scoring criterion.

The evaluation framework listed in the presentation contains the following measures:

- Accuracy: the proportion of evaluated samples classified correctly.
- Precision: among samples predicted as a class, the proportion that truly belongs to that class.
- Recall: among samples belonging to a class, the proportion correctly identified.
- F1-score: the harmonic mean of precision and recall.
- Confusion matrix: a table comparing actual and predicted classes.

Reported result

The PowerPoint reports Random Forest as the best-performing final model, with 89.20% overall test accuracy. It does not provide the Decision Tree's exact test accuracy, a numerical confusion matrix, or the complete precision, recall, and F1-score values. Those values should be inserted from the actual evaluation output if available. Because the CSV has many duplicate rows and deduplication changes the class distribution, confirm that the reported 89.20% accuracy was calculated using the same dataset version and preprocessing pipeline described in the final report.

Item	Reported information	Status
Models compared	Decision Tree and Random Forest	Specified in PPT
Validation/tuning	5-fold cross-validation and GridSearchCV	Specified in PPT
Best model	Random Forest	Specified in PPT
Test accuracy	89.20%	Reported in PPT
Decision Tree test accuracy	Not supplied	Verify from implementation
Precision / Recall / F1	Evaluation metrics listed; exact values not supplied	Add actual output
Confusion matrix	Listed as an evaluation method; matrix values not supplied	Add actual output

4.4 Implementation and Prediction Interface

The implementation begins by loading the soil dataset and preparing the input features and target label. Duplicate records are removed as described in the project presentation. Decision Tree and Random Forest classifiers are then trained and evaluated. Five-fold cross-validation and GridSearchCV are used for validation and tuning, after which the model results are compared.

The selected model is saved using Joblib. A Gradio interface provides fields for the 12 soil measurements. The user enters the values and submits the sample for prediction. The interface passes the input to the saved model and displays the predicted fertility class, with confidence or class probabilities where available.

24. Load the dataset.
25. Clean the data and remove duplicates.
26. Separate the 12 input features from the target fertility class.
27. Train the Decision Tree and Random Forest classifiers.
28. Validate and tune models using five-fold cross-validation and GridSearchCV.
29. Evaluate and compare the classifiers.
30. Save the selected Random Forest model using Joblib.
31. Load the model in the Gradio interface and predict the class for a new sample.

5. Conclusion

This project presents a machine-learning approach for soil fertility classification using 12 nutrient and chemical measurements: N, P, K, pH, EC, OC, S, Zn, Fe, Cu, Mn, and B. The workflow covers dataset preparation, exploratory analysis, model development, validation, evaluation, and a prediction interface.

Decision Tree and Random Forest classifiers are included to compare a single interpretable tree with an ensemble of trees.

According to the supplied project presentation, Random Forest achieved the highest reported test accuracy of 89.20% and was selected as the final model. Five-fold cross-validation and GridSearchCV form part of the validation and tuning workflow. The trained model is saved using Joblib and made available through a Gradio interface, where a user can submit a new soil sample and view the predicted fertility class.

The project demonstrates the potential of machine learning to support consistent and data-driven soil classification. However, the reliability of the prediction depends on the dataset and its labels. The system should be treated as a preliminary decision-support tool rather than a replacement for laboratory testing or professional agricultural advice. Future improvements may include using a larger and more diverse dataset, documenting the dataset source and class definitions, reporting complete per-class evaluation results, adding model explanations, and testing the system on independent soil samples.

Signature of the Team Member: _____